

v1.0Electron Neutrino Deep Dive

V1.0 to V2.0 Versioning Delta — Tardigradia SubParticles Engine

Architecture: One-Electron-Neutrino Universe Architecture **Type:** Scientific Change Log | **Date:** June 2026 **Applies to:** v1.0Electron Neutrino Deep Dive V2.0.md

Revision Necessity — Four Drivers

1. **Experimental discoveries 2022-2026** — LHC Run 3, LIGO O4, KATRIN 2024, NIF ignition, and other landmark results supersede V1.0 physics parameters
2. **New confirmed particles and phenomena** — Pines' Demon (2023), HH production evidence (2024), IceCube tau neutrino (2023), and others add entirely new simulation modes
3. **Platform upgrade** — WebGPU (Chrome 113+ 2023) replaces WebGL; enables 10^7+ particles
4. **Precision improvements** — CODATA 2022 constants, FLAG 2024 lattice QCD, NuFIT 5.3 oscillations

V1.0 vs V2.0 Specific Changes

Parameter	V1.0	V2.0	Severity	Consequence if Not Updated
KATRIN mass bound	$m(\nu_e) < 0.8 \text{ eV}$ (KATRIN 2022)	KATRIN 2024: $m(\nu_e) < 0.45 \text{ eV}$ (90% CL) — 44% tighter than V1.0	HIGH	Core parameter needs direct update to 2024 result
CNO solar neutrinos	pp chain only	Borexino 2023: CNO neutrino flux measured directly. 1.5% solar luminosity from CNO. Flux = $6.7 \pm 0.8 \times 10^8 \text{ cm}^{-2} \text{ s}^{-1}$	MEDIUM	New solar neutrino flux source for CNO cycle now available
Neutrinoless double beta decay	Mentioned conceptually	KamLAND-Zen 2024: $T_{1/2} > 3.5 \times 10^{26} \text{ yr}$. V2.0: $0\nu\beta\beta$ event = Majorana self-collision worldline. Currently unobserved	MEDIUM	Failure to observe $0\nu\beta\beta$ constrains Majorana mass — affects worldline topology choice

Parameter	V1.0	V2.0	Severity	Consequence if Not Updated
Reactor anti-ν_e measurements	General reactor context	SNO+ 2023: geoneutrino flux consistent with Earth composition models. New antineutrino flux map added	LOW	Geoneutrino source provides new ν_e flux in simulation

Simulation Impact Summary

The changes above drive the following modifications to the game engine architecture:

- **New particle types:** KATRIN
 - **Parameter updates:** All values in `static constexpr` declarations refreshed from 2022-2026 data
 - **WebGPU migration:** Particle update loop moves from CPU JavaScript to WGSL compute shaders
 - **New interaction modes:** Triggered by new experimental discoveries
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Unchanged Architecture

All foundational philosophy is **carried forward unchanged**: - Worldline Monism (One-Particle Universe) ontology - Proper time τ as primary particle identifier - Intrinsic vs. Extrinsic property distinction - Benevolence metric (C·F·S product formula) - Object-Oriented data structure (struct ParticleInstance extends to V2.0)

End of Versioning Delta | v1.0 Electron Neutrino Deep Dive V1.0 archived | V2.0 is the active specification as of June 2026